Principal Component Analysis

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Principal component analysis (PCA) is one of a number of exploratory statistical techniques that can be employed to recognize, interpret, and present patterns in multivariate datasets. PCA is ideal for investigating relationships between variables and cases, as well as characterizing significant data trends that involve multiple variables (Baxter 1994; Madsen 1988; Shennan 1997). Generally, PCA is ideal for exploring continuous measurement-based data associated with artifact morphology, elemental composition, and, as shown in this entry, spatiotemporal contexts. Below we outline how PCA works as a method of analysis and summarize two published cases that use this method to elucidate archaeological patterning.

For the sake of introduction, imagine a dataset consisting of values for 14 different measurements taken from a sample of stone projectile points (blade length, blade width, stem length, stem width, thickness, etc.). A perfectly accurate representation of the relationships among the variables and cases in this dataset would require 14 dimensions; however, humans can only simultaneously visualize up to three variables at a time. PCA and related techniques, including correspondence analysis and multidimensional scaling, allow analysts to overcome this dimensional limitation through ordination—a method that reduces the number of dimensions required to represent the many relationships among cases and variables in a dataset.

PCA achieves this dimensional reduction by grouping together variables in the dataset that are correlated. In most archaeological applications, covariance is expressed as a matrix of standardized correlation coefficients (r) for all variables, with values ranging between −1.0 and 1.0 (Baxter 1994, 65–66). PCA aggregates highly correlated variables together into summary variables called principal components (PCs). While each principal component is comprised of correlated variables, PCA requires that the principal components themselves be uncorrelated. When represented visually, these uncorrelated principal components can be seen as perpendicular (or orthogonal) axes that define a new space within which the variables and cases can be plotted with minimal loss of the original information (Shennan 1997, 281–282). PCA then determines new positions for variables and cases within this transformed space.

The technique provides powerful tools for exploring the relationships between variables and cases through a combination of numeric and visual results. There are three important and related types of numeric information calculated for each principal component—eigenvalues, component loadings, and component scores. An eigenvalue (or latent root) is most easily conceived of as the total amount of variation in a dataset accounted for by a particular principal component (Shennan 1997, 279). Eigenvalues themselves are difficult to interpret directly. Therefore, they must be transformed by dividing each value by the number of variables in the dataset and then multiplying by 100. The resulting value is the percentage of total variation in a dataset explained by that component. It is always the case that the first principal component will have the highest value, and subsequent components will be defined by decreasing eigenvalues (Baxter 1994, 67).

PCA results typically include a table listing component loadings for variables (rows) and principal components (columns). Component loadings are standardized measures of covariation between each principal component and each variable in a dataset and can be interpreted exactly like correlation coefficients (r) (Shennan 1997, 278). The analyst can determine which variables are most closely related to a particular

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principal component by identifying the variables with strong positive or negative loadings. By squaring component loadings, the analyst can also determine what percentage of variation in each variable is explained by a principal component. These two procedures are essential for identifying and interpreting the data trends that are being marked by the principal components.

PCA also provides a table of numeric results called component scores relating cases (rows) to principal components (columns). Component scores can be thought of as translated coordinate values marking the position of a case along each of the principal components. Each component score is derived from the original variables by multiplying the standardized value of each case by the component loading of each variable and summing the results for all of the variables (see Shennan 1997, 285–187 for a detailed discussion). Analysts can best understand the data trends captured by principal components by examining cases with component scores at both the positive and negative ends of a principal component.

While the numeric results of PCA are crucial for identifying and interpreting patterns in multivariate datasets, the visual results of PCA are perhaps better suited for presenting those patterns to an audience. Visual representation of data is often the main goal of ordination methods like PCA; these are scatterplots that depict the relationships between variables and cases along axes representing the principal components. Typically, PCA results are presented as separate scatterplots for variables and cases, respectively. Because component loadings are used as coordinates for variables and component scores for cases, the scatterplots distill much of the data contained in the numeric results; spatial clustering of variables or cases suggests similarities. Additionally, because both cases and variables are plotted within the same relative principal component dimensions, the relationships between variables and cases can be represented spatially (Baxter 1994, 57–59). Looking at the paired scatterplots, *variables* occupying the same relative principal component space as a cluster of *cases* can be said to have significant influence over the observed similarities among cases in the cluster (Shennan 1997, 295–296). Scatterplots also allow the analyst the additional option of coding cases with different symbols to represent other dimensions, such as time or space, which may aid in further understanding the data trends.

To demonstrate the utility of PCA with respect to archaeological data, we present two cases from the published literature. The first case highlights the efficacy of using PCA to quantitatively integrate seemingly disparate types of data: animal bone fragments and carbonized plant remains. Most often these two types of data are analyzed separately, which can create an artificial conceptual divide between the meat and vegetable portions of the diet (VanDerwarker and Peres 2010).

In their case study from Formative-period Veracruz, Peres, VanDerwarker, and Pool (2010) consider how sociopolitical status and contexts of discard vary with respect to plant and animal foodways at the multi-mound political center of Tres Zapotes. Based on architectural and artifactual analyses, Peres and coauthors were able to define three broad types of socioeconomic space: nonelite domestic space, elite households, and ceremonial areas. In analyzing how sociopolitical space varies in relation to food use, the authors begin by conducting independent analyses of the archaeobotanical and zooarchaeological datasets. Independent analysis of the plant data did not reveal much difference in plant presence or abundance between these three contexts; however, beans appear to have been restricted to elite contexts, and nonelite contexts appear to have had elevated levels of tree fruit remains. Independent analysis of the faunal data revealed a greater concentration of deer and dog remains in elite contexts versus fish and turtle remains in nonelite contexts. Patterning in ceremonial areas was less clear when compared to the elite and nonelite household areas.

In order to clarify the patterning in ceremonial areas, and in line with the idea that ancient food practices should be conceptualized as cuisine in which ingredients are combined to form distinct dishes, Peres et al. (2010) turned to PCA. To make the plant and animal datasets comparable, each dataset was standardized in a similar manner. Frequencies of plant fragments were summed by type (maize, fruits) for each context (nonelite, elite, ceremonial), and each sum was then divided by the total plant weight in that context. Likewise, animal bone frequencies were also summed by type (deer, dog, fish, turtle), and divided by total bone weight for each context. The resulting continuous data were then used in the PCA. The integration of these datasets revealed patterning that was not apparent through the independent analyses alone. Both plant types (maize, fruits) mapped directly onto nonelite domestic contexts, with values that did not overlap with the elite or ceremonial contexts. Given that maize was present in similar frequencies across all three contexts in the independent analysis, the fact that the PCA results associate maize specifically with nonelite contexts suggests that plant food frequencies in elite and ceremonial contexts were overwhelmed by a much higher frequency of meat resources. Thus, the PCA reveals that the nonelite diet was focused almost entirely around plant foods, with meats being preferentially incorporated into elite diets and ceremonial events. Such an interpretation would have been impossible without integrating these datasets using PCA. While this case employs subsistence data, any type of material data could be incorporated, with the caveat that all data used in the PCA are transformed into continuous data and standardized in similar ways to make them comparable.

PCA is also useful when trying to determine spatial clustering of multiple variables. In order to be appropriate for this type of analysis, the cases must represent discrete spatial contexts (e.g., features, units, houses). The case study we use to exemplify PCA as a tool of spatial analysis is a burned house structure from the Ravensford site (ca. 1700–1830 CE), located in western North Carolina (VanDerwarker, Alvarado, and Webb 2014). The structure was excavated in a series of contiguous 1×1-m units, totaling 69 units. Given the details of timing, burning, and house abandonment, the plants recovered from the floor of this structure relate to activities that occurred within it toward the end of its use-life. PCA was thus used to determine how the structure's ancient inhabitants organized food storage and processing tasks.

VanDerwarker and colleagues (2014) use the 1×1-m excavation unit as the spatial unit, in which each unit represents a single case in the analysis. The variables are represented by the plants that were identified in those units, and all plant frequencies were standardized to total plant weight for each respective excavation unit. The first PCA run identified several units that formed two distinct clusters. The first cluster was represented by a single unit located near the southern corner of the structure. The second cluster included five contiguous units in the northwestern area of the structure. In order to determine additional patterning, a second PCA was conducted after excluding the units from clusters identified in the first analysis. The second run resulted in one additional unit being included in cluster 1, one additional unit being included in cluster 2, and a third unit creating a new cluster (cluster 3).

Based on the spatial clusters defined through the PCA, the authors were able to interpret how interior food storage and processing were organized within the house. Cluster 1 was composed mostly of nutshell and maize remains, suggesting that the southern corner of the structure was an area dedicated to staple food processing. Cluster 2, located in the northern corner of the structure, was dominated by fruit seeds and interpreted as an area of dried fruit storage. The authors suggest that fruits were dried and then placed in storage by hanging them from ceiling rafters. When the house burned and the roof collapsed, these fruits likely fell to the floor amid a background of general food-processing debris. Cluster 3, in the west-central area of the structure, was dominated by bottle gourd rind and seed fragments. Bottle gourds were primarily used as liquid storage containers and were probably suspended from the rafters, much like the dried fruits. When the roof beams collapsed during the structure's incineration, the gourds would have crashed to the floor and shattered, littering the floor with a high density of bottle gourd rind fragments. This example demonstrates how PCA can be employed to define clear activity clusters in situations where material remains represent abandonment debris or primary refuse.

Complex multivariate statistics like PCA can seem daunting to the quantitatively challenged archaeologist, but PCA's utility in dealing with large datasets to reveal novel patterning is clear. The cases above exemplify the interpretative power of PCA to move us beyond what we can learn from simple univariate analyses. PCA is thus an important technique to curate in our toolkit of archaeological data analysis.

SEE ALSO: Cluster Analysis; Compositional Data Analysis; Computer Applications in Archaeology; Correspondence Analysis; Factor Analysis; INAA and Material Analysis; Inferential Statistics; Instrumental Analysis of Lithics; Neutron Activation Analysis (NAA); Statistics in Archaeology

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